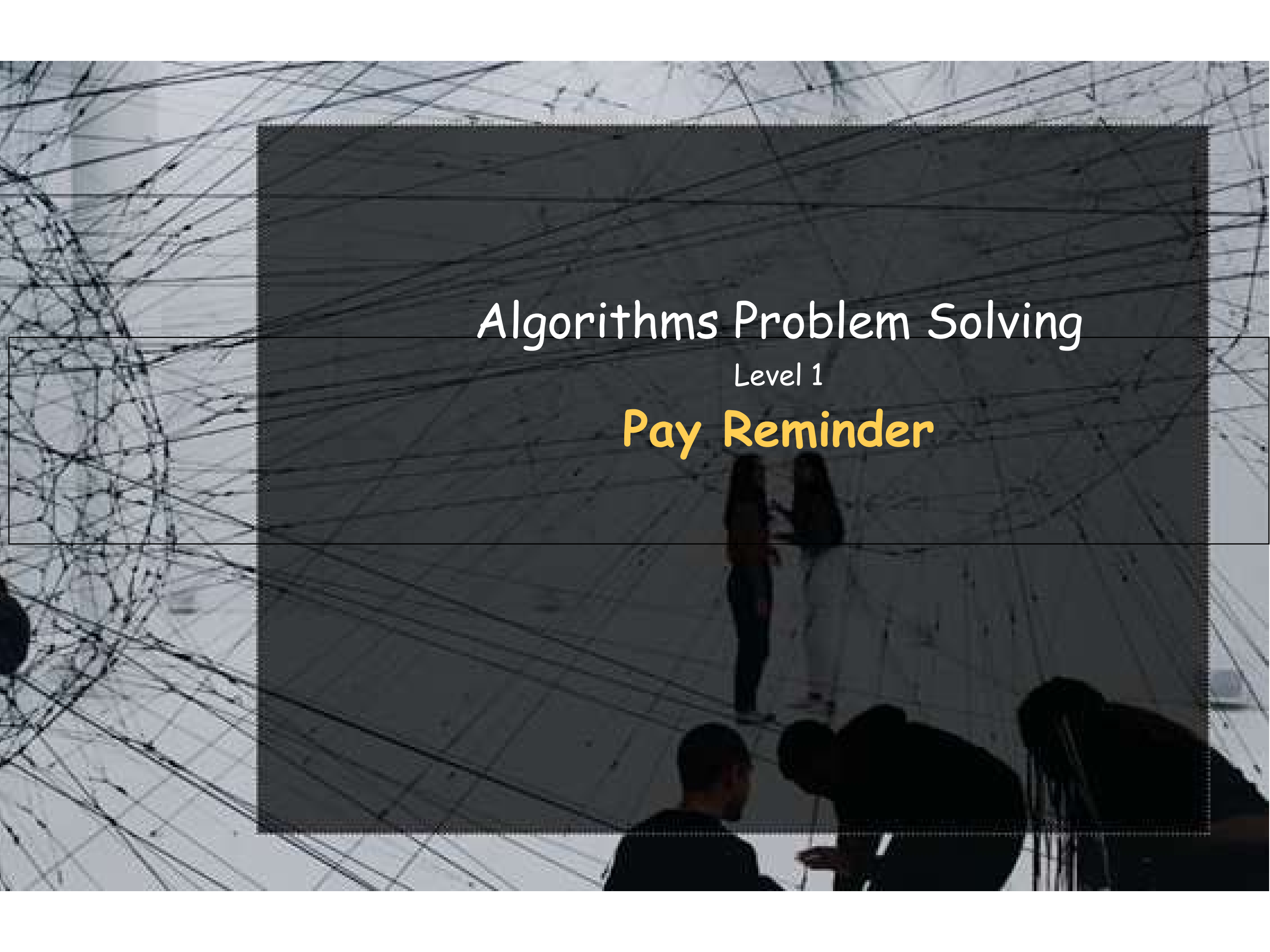


Algorithms Problem Solving

Level 1

Pay Reminder

The background of the slide is a photograph of a modern building's interior. It features a large, curved glass ceiling that allows natural light to filter through, creating a pattern of light and shadow on the floor. In the foreground, the silhouettes of several people are visible, some standing and some sitting, suggesting a collaborative workspace or a public area. The overall tone is professional and contemporary.

Algorithms Problem Solving

Level 1

Pay Reminder

Definition

- ◆ This program reads the total bill and the amount of cash paid by the customer, then calculates the remaining amount to be returned.
- ◆ The calculation is:
$$\text{CashPaid} - \text{TotalBill}$$
- ◆ The program uses a structure, functions, and returned values.

Explanation

- ◆ The struct `stBill` groups the related bill information:

TotalBill stores the total cost.

CashPaid stores the amount paid by the customer.

- ◆ `readInput()` reads both values and returns them as an `stBill` object.

- ◆ `calculateReminder()` subtracts the total bill from the cash paid.

- ◆ If the customer pays more than the bill, the result is the amount of change to return.

- ◆ `printReminder()` displays the calculated result.

- ◆ In `main()`, the structure is passed through the calculation function, and the result is passed to the printing function.

Steps in main()

1. Create the `stBill` structure.
2. Read the total bill.
3. Read the cash paid.
4. Return the bill information.
5. Calculate `CashPaid - TotalBill`.
6. Return the calculated remainder.
7. Print the result.

💡 Example

◆ If the user enters:

Total Bill: 20

Cash Paid: 50

◆ The calculation is:

$$50 - 20 = 30$$

➡ Result

The Remainder = 30

Note

- ◆ The program assumes that `CashPaid` is greater than or equal to `TotalBill`.
- ◆ If the customer pays exactly the bill amount, the remainder is ``0``.

⊘ Warning

◆ If `CashPaid` is less than `TotalBill`, the function returns a negative value. The program does not currently handle insufficient payment separately.



Comparison

▨ CashPaid > TotalBill

- ✓ The customer paid more than the bill.
- ✓ The difference is returned as change.
- ✓ Example: $50 - 20 = 30$.

▤ CashPaid < TotalBill

- ✓ The customer did not pay enough.
- ✓ The result is negative.
- ✓ The program does not display a specific insufficient-payment message.

✦ Summary

- ✓ A structure can group related bill information.
- ✓ The remainder is calculated using `CashPaid - TotalBill`.
- ✓ Functions separate input, calculation, and output.
- ✓ A negative result indicates that the cash paid is less than the total bill.

Thank you 😊